Testing the Test Oracle: An Evaluation Harness for Meaning Preservation in LLM-Expanded AAC Text

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***Abstract*—Developers of LLM keyword-to-sentence expansion for augmentative and alternative communication (AAC) need tests for whether a person's meaning survives, but the test oracle, an automated checker of what the output says, is itself software that can be wrong. We present a small evaluation harness built around that problem: five input-conditioned requirements that score a choice only when the input carried it, minimal-pair families rendered through five input channels, an LLM-extractor oracle whose verdicts are computed in code, and tests of the oracle itself. On 547 human-written pairs from 20 writers (an exploratory replication, not preregistered, with 6–7 writers per contrast), four models gave the affirmative reading on 99.8% of bare polarity outputs, a speech-act tag raised preservation by 35.3 pp over a length-matched word [31.2, 40.3], and fully typed plain requests gained a politeness marker in 7.9% of outputs against 0.3% removals, driven mainly by one model. The direction of all three results held on a machine-written confirmatory set whose contrasts were pre-specified and git-timestamped. Testing the oracle mattered: relaxing a rule to remove a false alarm hid real substitutions, and an input-builder defect (7/139 restorations) was found by auditing against an executable requirement, which now rejects it before generation. The oracle's critical labels agreed with human coders at $\kappa = 0.97$ and 0.94 on two new sets of outputs.**

***Index Terms*—accessibility, AAC, LLM testing, test oracle, requirements, regression testing**

I. Introduction

Many people who use AAC with letter or word selection compose a message from a few keywords, because each selection costs time and effort [1], [2]; LLM systems expand such a selection into a sentence [3], [2]. Symbol- and grid-based AAC is out of scope here. A test for such a system must decide whether an edit is a defect, and that depends on what the input carried. From the keywords `supervisor read` no model can know whether the supervisor does *not* read; the affirmative guess is a default, not a violation (Fig. 1, card 1). When a person supplies a cue, or a whole sentence, the output must keep it, and a changed value is a violation.

The software-engineering problem is the oracle. Deciding whether an output "kept the meaning" needs a checker, here an LLM that extracts a few explicit labels, and an unvalidated checker silently decides every result. We therefore treat *testing the oracle* as a first-class contribution beside testing the models. We contribute: (1) a harness with five requirements R1–R5 that have stated sources and executable tests (§II); (2) tests of the oracle, including two defects we introduced ourselves, each traced from symptom to regression test (§IV); (3) evidence from a human-written replication (hw1, primary) and a machine-written confirmatory set (W-C) that these tests find real, model-specific behavior (§III); and (4) the harness and stored outputs as an artifact. The harness is for developers; it is not a tool for assessing AAC users or their messages. We ran no study with AAC users and make no claim about benefit, effort or preference.

II. Requirements and Harness

Families and channels: A *family* is a minimal pair of source messages that share content words and differ on one *critical* dimension: polarity, agent (who acts), certainty, speech act (question, request or statement), politeness, or time/quantity. Each variant is rendered through five channels: *bare* keywords; +*word*, which adds one noncritical content word; +*chip*, which adds the critical cue as a bracketed tag (`[NOT]`); +*all*, with every tag the variant needs; and *full*, the source sentence itself. Every variant needs at least one tag, so the unmarked reading gets a counter-tag (e.g., `[PLAIN]`). A chip is a *probe* of whether supplied information is used, not a proposed interface: its real cost depends on the access method (a dwell, a menu traversal, a grid button), and it adds a vocabulary to learn and an error-recovery path.

Requirements: For output y from input x, let $L_d(y)$ be the extracted value on dimension d and $C(x)$ the commitments that x carries. Table I states each requirement with its source and its test. A *Violation* means a supplied value was not kept (R1, R2); an *Assumption* means the output committed to a value the input left open (R3). A reference-fidelity score would instead charge the model for what the interface omitted, and a similarity score would pass an added "please" on a typed sentence. All five requirements are our proposal, derived from the structure of the task and from defects we met; none was derived with AAC users or clinicians. The declared policy behind R2, do not change typed wording unless the person asks for rewriting, protects authorship; some users prefer that a system rewrite their text, so R2 is a policy to test, not a recommendation.

Harness: Fig. 2 shows the pipeline. A family specification is rendered into channels; the R5 validator checks that

	Exact stored input ⇒ stored output (judged label)	What it shows, and who would own the fix	Rate in the example's stratum (hw1, prompt P0, four models)
1 polarity Gemma-3-4B, writer P18; target *negated*	bare `Keywords: supervisor read` ⇒ "The supervisor read." (affirmative) +chip `Keywords: [NOT] supervisor read` ⇒ "The supervisor did not read." (negated) full `The user typed: The supervisor does not read my schedule.` ⇒ "The supervisor does not read my schedule." (negated) *hidden source:* "The supervisor does not read my schedule."	A bare input does not say whether the sentence is negated, so the affirmative is an *Assumption*, not a model error. With the supplied cue or the typed sentence, negation is kept. *Owner of the fix (our taxonomy's choice):* the input channel.	bare commits to a wrong value: 49.8% [49.5, 50.0]% of bare rows (7 writers); then the `[chip]` output matches: 99.3% [98.6, 99.8]% (414). First of 104 qualifying families by hash.
2 speech act Gemma-3-4B, writer P01; target *question*	bare `Keywords: cafe lunch` ⇒ "I want a lunch at the cafe." (statement) +chip `Keywords: [?] cafe lunch` ⇒ "Is the cafe open for lunch?" (question) full `The user typed: Does the café serve lunch near the station?` ⇒ "Does the café serve lunch near the station?" (question) *hidden source:* "Does the café serve lunch near the station?"	A bare input leaves question vs. statement open; the statement is an *Assumption*. With `[?]` this output is a question, but a supplied tag is not always kept. *Owner:* the input channel for the bare miss, the model when a supplied tag is dropped.	bare commits to a wrong value: 56.0% [52.1, 63.0]% of bare rows (6 writers); then the `[chip]` output matches: 75.3% [66.0, 83.2]% (300). First of 66 qualifying families by hash.
3 politeness gpt-6.1-terra, writer P04; target *plain*	bare `Keywords: save seat window` ⇒ "Please save me a seat by the window." (polite) +chip `Keywords: [PLAIN] save seat window` ⇒ "Save me a seat by the window." (plain) full `The user typed: Save my seat near the classroom window.` ⇒ "Save my seat near the classroom window." (plain) *hidden source:* "Save my seat near the classroom window."	Bare keywords carry no politeness marker, so "Please" is an *Assumption*; with the typed sentence the plain wording is kept. *Owner:* the input channel.	bare commits to a wrong value: 50.0% [50.0, 50.0]% of bare rows (6 writers); then the `[chip]` output matches: 82.0% [79.5, 84.6]% (356). First of 89 qualifying families by hash.

Fig. 1. Selected stored outputs on *human-written* pairs, verbatim; each input is the exact string sent to the model, wrapper included. Selection is by a declared rule: per dimension, the first family in SHA-256 order of its ID with a bare output that commits to a wrong value while the chip output for the same family, variant and model is right. The examples are therefore not typical, and the right column gives the rate in each stratum. For a binary dimension the bare rate is near 50% by design, so it is not a finding. The "owner" labels are a design choice of our taxonomy, not an empirical result. Source: `analysis/hw1/examples.md`.

each displayed cue matches the claim; a generator produces outputs; an extractor (gpt-6.1-sol, a different model from the one under test) answers the family's critical question and five general questions from the output alone, with added-content extraction also seeing the visible input. Two extractor prompts (J1, J2) differ in wording and option order; critical disagreements go to a cross-family arbiter (claude-opus-5-5) or stay unknown. Verdicts are then computed in code from labels, item specification and channel, so a rule change re-scores every stored output without model calls. The oracle decides only whether a fixed set of explicit, text-level values is kept; it does not judge fluency, referents or whether anyone would accept the sentence.

Reuse: A developer adds a requirement as a function of labels, item specification and channel, re-scores stored outputs without any model call, and runs the identity and keyed-mutant tests before trusting the new rule. Re-scoring and the oracle tests need no API access; regenerating outputs needs the API models, whose versions may be retired and whose outputs are not bit-reproducible.

Evidence populations: All items are authored or crowd-written, not messages from AAC users. *hw1* (primary): 547 pairs kept of 574 candidates, written by 20 non-author writers over six dimensions, each kept after three blind independent checks; 546 families are analysed. *W-C* (confirmatory, machine-written): 238 fresh families on three dimensions, whose three contrasts were pre-specified in a git-timestamped plan frozen before generation. *Discovery*: 60 authored families, used to develop prompts and codebooks. *W-B*: crowd-written fictional messages from AACText [4], used for corpus transfer (§IV). Four models run with a single one-sentence prompt (P0, greedy): Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it, three open models run locally with 4-bit quantization, and one API model, gpt-6.1-terra. All intervals are pointwise 95% percentile intervals from writer-cluster (hw1) or seed/construction-cluster (W-C) bootstraps, with 10,000 resamples over a fixed model roster.

III. Results: Does the Harness Find Real Behavior?

A. Primary replication on human-written pairs (hw1)

Table II gives the three contrasts, which we name *Act-gain* (speech-act chip minus a length-matched word), *Politeness* (politeness added to plain typed sentences minus removed from polite ones) and *Default* (share of bare polarity outputs that are affirmative). They were pre-specified for W-C as P1–P3 (unrelated to prompts); in hw1 they are a replication and were not preregistered. All hold under Holm correction, but each rests on only 6–7 writers, so we claim direction and consistency, not interval width; a writer-cluster percentile interval with so few clusters is probably too narrow. The chip gain is positive on all six dimensions, between 35.3 and 82.7 pp, with every lower bound above 31.2 pp. Collapsing near-duplicate pairs, excluding one writer, reversing the arbiter's judgment

TABLE I

TRACEABILITY OF REQUIREMENTS R1–R5: WHERE EACH COMES FROM, WHICH TEST CHECKS IT, AND WHAT THAT TEST HAS FOUND. ALL CHECKS RUN ON STORED OUTPUTS AND RE-SCORE WITHOUT MODEL CALLS.

Requirement	Source	Executable test	What the test has found
R1 supplied cue: if $C(x)$ fixes d to c, $L_d(y)$ must be admissible for c	Interface contract: a cue the person supplied is information the person paid for	Cue check on *+chip*/*+all* outputs; Violation if the value differs	Speech-act tags kept in 73.24% of W-C targets vs. 99.20% for the typed sentence; 58 of 208 `[?]` targets not kept
R2 full input: each scored explicit value of a typed sentence is kept, including the absence of a politeness marker	Declared policy (above); exact copy satisfies it	Full-input check on *full* outputs	Politeness added to plain typed sentences: 7.9% (hw1), 8.12% (W-C)
R3 ambiguity: if $C(x)$ leaves d open, $L_d(y)$ is not pass/fail; log an Assumption and report its direction	Bare keywords under-specify; a systematic default decides which reading is free	Default report on *bare*/*+word* outputs; 50% paired match is a design ceiling	Affirmative reading on 99.8% (hw1) and 99.37% (W-C) of bare polarity outputs
R4 composition: adding a compatible cue for e must not break a kept value on $d \neq e$	Cues are supplied together in practice	Composition check: pass→fail transitions of the protected value, minus the repeat-draw floor	One exploratory study (W-D), omitted here: Llama lost a kept negation in 55/640 transitions
R5 input builder: a test input that claims to supply c must display a cue for c	Defect B below: failures cannot be attributed to a model if the input lied	Validator on the rendered input (`validate_cue_inputs.py`), run before generation	7/139 restorations contradicted their source; the validator rejects 7/7

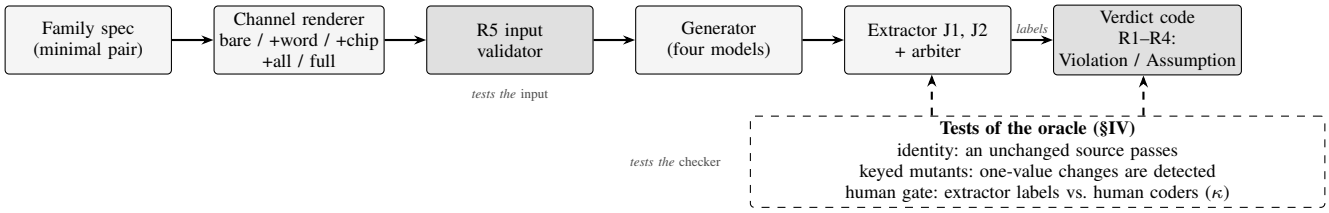


Fig. 2. Evaluation harness. Solid arrows: data flow for one run. Dashed box: tests of the oracle itself (§IV). Shaded boxes: the code-level components that enforce R5 and R1–R4. Because verdicts are code over stored labels, a rule change re-scores every output without model calls.

TABLE II

HW1 (PRIMARY, HUMAN-WRITTEN, 546 FAMILIES) AND W-C (CONFIRMATORY, MACHINE-WRITTEN, 238 FAMILIES): THE THREE CONTRASTS, PROMPT P0, FOUR MODELS, 95% CLUSTER-BOOTSTRAP INTERVALS, HOLM OVER THE THREE. HW1 WRITERS PER CONTRAST IN PARENTHESES. PRE-SPECIFIED NAMES: ACT-GAIN = P1, POLITENESS = P2, DEFAULT = P3.

Contrast	hw1 [95% CI] (writers)	W-C [95% CI]
Act-gain, pp	+35.3 [31.2, 40.3] (6)	+31.41 [25.00, 37.94]
Politeness, pp	+7.6 [5.0, 10.4] (6)	+6.25 [2.50, 10.00]
Default, % affirmative	99.8 [99.3, 100.0]% (7)	99.37 [98.42, 100.00]%
Holm p	.0003 (all three)	.0003 (all three)

calls or using all candidates flips no decision; the largest shift is 1.7 pp (4.4 pp with all candidates). Because the chip text enters the model's prompt, Act-gain and Default are close to manipulation checks of whether the harness separates supplied from missing information, which is their purpose.

B. Confirmatory machine-written set (W-C)

The same three contrasts hold on W-C with the same direction and a similar or smaller size (Table II). Confirmation is qualified. First, 48 cached requests predate the final item freeze. Second, Act-gain relies on post-hoc arbitration of judge disagreements (259 rows); with those labels masked as unknown it is 31.65 pp [25.22, 38.19], a sensitivity analysis, and Politeness and Default contain no arbitrated rows. "Pre-specified and git-timestamped" means the plan was committed to a private repository before generation; it is not independently verifiable. A hash of the protocol was deposited publicly as GitHub release `protocol-hash-2026-11-01`; it post-dates the W-C freeze and so does not certify it.

C. Assumptions, full-input violations and model heterogeneity

Fig. 4 separates the two kinds of evidence. Under bare input, 317/319 directional polarity outputs in W-C were affirmative; these are Assumptions, not failures, and the interface decides who pays an extra selection to say "not". That default may be an artifact of heavy compression, so we asked humans: 20 annotators typed keywords for 600 crowd-written AACText messages (1,800 sets); on the 122 messages with a negation cue they dropped it in 20.8% of sets, against 100.0% for a deliberately crude four-word rule-based compressor, so the Assumption channel is realistic but the compressor is not a user model. Full-input violations are model-specific: in hw1 the Politeness contrast is 22.5% of outputs for Gemma, 6.7%

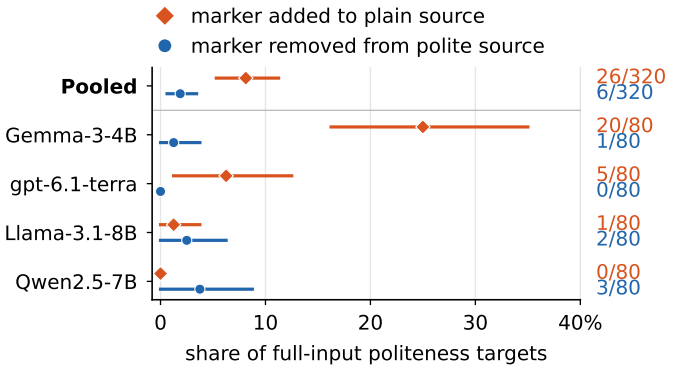


Fig. 3. W-C, full input (the source sentence is visible to the model), P0, 80 politeness families: politeness marker added to a plain source vs. removed from a polite one, per model and pooled; 95% cluster-bootstrap intervals; counts at right.

for gpt-6.1-terra, 1.1% for Llama and 0.0% for Qwen (net additions minus removals); in W-C, plain typed sentences gained a politeness marker in 26/320 outputs, driven by Gemma (20/80) and gpt-6.1-terra (5/80), while Llama and Qwen removed markers more often than they added them. This is a regression to test for per model, not a property of LLMs (Fig. 3). Question tags are the weakest supplied cue: one tag kept the act in 73.24% of W-C targets, and 42 of the 58 failures were coded as requests ("Can you…?"), a codebook decision that we flag rather than resolve.

IV. Testing the Oracle

Every result above is only as good as the checker, so we test it as software, in the spirit of the test-oracle literature [5], [6] and of mutation analysis [7]: an authorized change must pass and an unauthorized one must fail. *Identity test:* an unchanged source injected as its own full-channel output must pass. *Keyed mutants:* a hand-written one-value change, with keys hashed before the first judge call, must be detected. *Human gate:* extractor labels against human coders.

Two defects, two regression tests: Table III traces the two defects we introduced ourselves. In Defect A, the identity test failed 14 of 2,292 reproductions, all one sentence, "Yes, you can borrow my car this weekend.", coded *statement* against the target *acceptance*. The model had done its job; the checker demanded a distinction the text need not force. The obvious repair, accepting *statement* for an acceptance target, passed identity but failed the next test: the relaxed oracle detected only 4/6 acceptance mutants and missed both statement substitutions ("You borrowed my car this weekend."), whereas without the relaxation it detected 24/24 mutants across groups. Removing a false alarm removed fault sensitivity, so both refusal/acceptance families are quarantined from every primary result (leaving 58 of 60 discovery families) until a predicate passes both tests. In Defect B, an audit of 139 restored-cue inputs built from AACText messages found 7 whose displayed chip contradicted the stored source, e.g., `[MAYBE]` for a plain assertion, after a silent fallback in our builder. In the 5 `[MAYBE]` messages, all 20 outputs followed the visible chip and failed the source, which a source-fidelity score counts as model failures. The R5 validator, which decodes only the rendered input, now rejects all 7 before generation, matches the audit on 139/139, and never certifies the 8 coarser `[SURE]` restorations.

TABLE III
Defect-to-regression-test trace.

	A: relaxed acceptance oracle	B: input-builder fallback
Symptom	unchanged source fails (14/2,292)	chip contradicts source (7/139 restorations)
Found by	identity test	audit against R5
Fix	accept *statement* as acceptance	decode rendered input; reject conflicts
Cost of fix	substitutions hidden (4/6)	none; coarser chips stay unknown
Test now in suite	keyed acceptance mutants; quarantine on failure	R5 validator before generation; replay agrees on 139/139

TABLE IV
Oracle evidence. Mutation and identity checks are on clean authored sentences (Wilson 95% intervals), not error rates on model outputs. Human rows compare extractor labels with a plurality of human coders.

Check	Result
Identity: unchanged source, false alarms	0/120
Scoped single-value mutations detected	120/120
Keyed challenge mutants detected (gate)	22/24
Keyed challenge controls, false alarms	0/24
acceptance group, relaxed rule	4/6
all groups, without the relaxation	24/24
Human gate, discovery items (187 items, two coders): held-out κ, 7 fields	0.91–0.97
Outputs for human-elicited keywords (4 coders, 151 items): critical κ [CI]	0.97 [0.94, 0.99]
verdict agreement	151/151
Outputs for hw1 pairs (3 coders, 150 items): critical κ [CI]	0.94 [0.89, 0.97]
verdict κ	0.81–0.85

Mutation and identity evidence: Table IV lists the evidence on authored inputs. The keyed challenge set has four groups of six controls and six one-feature mutants; its pre-specified gate ($\geq$22/24 mutants detected, $\leq$1/24 false alarms) passes when pooled, with no slack.

Human gate: The first human gate, with two annotators, was run on the *discovery* items (the extraction inventory used to develop the codebook), with plurality references and tied items excluded rather than adjudicated; its pre-specified criterion ($\kappa \geq 0.6$) passes on all 7 fields. It validates extraction of explicit markers on those items, not repaired text or W-B readings. Two later checks use outputs the harness had not been developed on: outputs for human-elicited keywords and outputs for hw1 pairs (Table IV). They are the oracle's evidence on the headline-type data, though their coders were also taught the codebook, and the verdict κ values on hw1 rest on small counts of positive verdicts. Machine agreement is not validation: on "I can't wait to go to the party.", both judge prompts, the rule-based extractor and two further

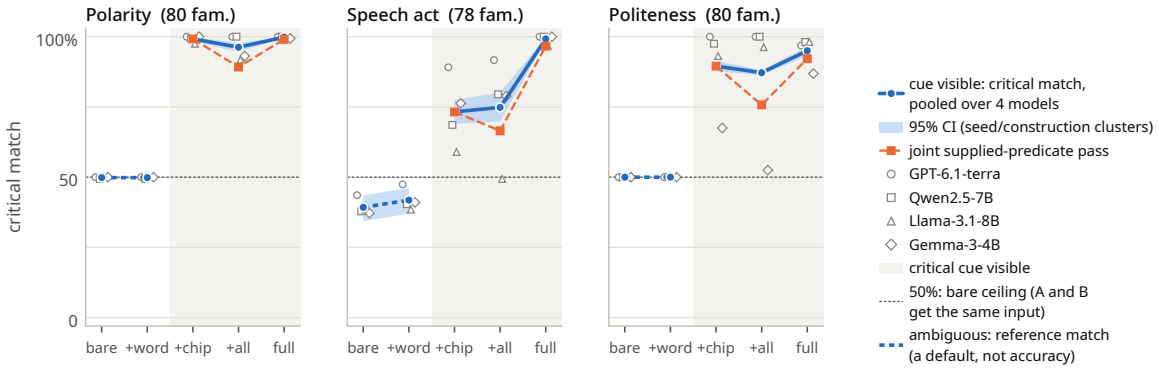


Fig. 4. W-C critical-value match by channel (P0, four models pooled; markers: individual models). Left of the shading, input is ambiguous: match is the chosen default, bounded by a 50% design ceiling, not accuracy. In the shaded region the cue is visible and match measures preservation. Orange squares also require every other supplied predicate to be kept.

machine coders answered *negated*, and both human annotators answered *affirmative*, as keyed.

V. Related Work

Test oracles, metamorphic testing and requirements for ML: Whether an output is correct is the oracle problem [6]; for programs without a practical oracle, partial or pseudo oracles are standard [5], and metamorphic testing checks relations between runs instead [8]. Mutation analysis measures a test suite by whether it detects seeded faults [7]; we apply that idea to the oracle, not the system. Requirements engineering for ML stresses that data-dependent behavior needs explicit, testable requirements [9]; our R1–R5 are one such set, with sources and tests stated.

Testing LLM systems: CheckList organizes behavioral tests into minimum-functionality, invariance and directional tests [10]; Cho et al. catalogue metamorphic relations for LLM-based NLP [11]; COLLIE tests compositional constraints [12]; Sensei combines a user simulator with assertions and metamorphic relations [13]. We claim no new testing technique. LLM judges show position, verbosity and self-preference biases [14], [15], and similarity metrics can miss negation [16]. Pauli et al. show that preservation metrics look valid when test data confound style change with content errors [17]; our identity and keyed-mutant tests apply that idea to a checker.

Preservation and LLMs in AAC: Qu et al. evaluate speech-act retention in style transfer [18]; rewriting can keep semantic similarity while erasing pragmatic markers [19]; and LLMs favor particular politeness strategies [20]. AAC systems studied with users expose intent controls [1], [21], [22], [23]; Lacosse et al. keep bracketed function words such as "not" [24]; Choudhury et al. test whether affect cues reach the output [25]; Gorenshtein et al. benchmark intent drift [26]; R-SPEAK scores faithfulness and was co-designed with people with aphasia [27]. We add a conditional specification of which commitments are testable under each input, and tests of the oracle that checks them. Accessibility engineering spans the life cycle [28], but conformance checks do not see whether generated text keeps a user's meaning.

VI. Threats to Validity

Construct. Labels are coarse, explicit categories; "plain" means no explicit marker, and referents, negation scope and certainty degree are only partly covered. Passing the checker is weaker than keeping the typed wording, and a scored repair is not necessarily a sentence someone could say; a small human audit of one companion repair method, not reported here, supports that caution. *Oracle.* All rates depend on an LLM extractor and arbiter; the arbiter ran after outcomes were known, and human coders were taught the codebook. The first human gate's κ is on discovery items, not on confirmatory data. *Internal.* W-C has early cached requests and post-hoc arbitration; hw1 is not preregistered and has few writers per contrast. *External.* Four models, three small and quantized; one prompt; English; constructed and crowd-written, not AAC-user, messages. Model versions may be retired, and API outputs are not bit-reproducible. *Conclusion.* Effects are model-specific, and zero observed events do not establish zero risk. Selection counts omit reading, navigation and fatigue [29].

VII. Conclusion

An oracle for meaning preservation is software and needs tests of its own. Ours caught a repair that hid real substitutions and a builder that supplied the wrong cue, and its labels agree with human coders on two new sets of outputs. With it, a human-written replication finds that bare keywords leave a default reading, that supplied speech-act cues help but are not always kept, and that some models change typed text. Families, requirements, harness, oracle tests, stored outputs and analysis code are available at https://github.com/1unarzDev/aac_voice_preservation.

Ethics and Positionality

No data were collected from AAC users; AACText is public under CC BY 4.0. All annotators were adult non-authors recruited from the authors' university classmates, identified only

by an ID; each consented and was reimbursed for their time (payment and food). They labelled model outputs and audited items under a fixed codebook, blinded to model, condition and source. Some also chose keywords for public AACText messages, and some wrote new sentence pairs, which they released under CC BY 4.0 with recorded consent. No personal or health data were collected. As independent researchers, we did not seek an institutional review determination. No AAC users, speech-language pathologists or autistic advisers took part in this work, which limits it: every requirement here is our assumption about what matters, not a finding about what AAC users want, and the requirements should be revisited with AAC users before any deployment decision. We drew on writing by and with autistic and AAC-using adults [30], [31], [29] and on participatory-research and double-empathy work [32], [33] to avoid encoding a group style or deficit; that reading is not a substitute for participation. We assume no "autistic style", and our claims are limited to commitments explicit in the text.

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